Foo Wei Ming

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Summary

Computer Science and Design undergraduate at SUTD (Merit Scholar) with a cybersecurity diploma, a machine learning and NLP internship at SAP, and RSAF experience spanning aircraft engineering and digital innovation.

Education

Singapore University of Technology and Design (SUTD) **Singapore**
B.Eng. Computer Science and Design (CSD); Undergraduate Merit Scholarship *Sep 2025 – 2029 (Expected)*

- cGPA: 4.22/5.00
- Data Driven World (Python, data structures, algorithms, OOP, linear and logistic regression); Modelling and Analysis (calculus, numerical methods)

Singapore Polytechnic **Singapore**
Diploma in Information Security Management (Cybersecurity) *Apr 2020 – Apr 2023*

- Relevant modules: Internship (Distinction), Programming for Data Science (Distinction), Secure Coding, Information Security, Linux Administration and Security, AWS Cloud Foundations, Programming in Security
- Final Year Project (Grade A): AlphaReGen, a network security simulation (see Projects)

Experience

Republic of Singapore Air Force (RSAF) **Singapore**
Developer, Agile Innovation Squad *Aug 2026 – Dec 2026*

- Supported digital transformation by migrating Excel-based tools to Microsoft Copilot Studio
- Debugged VBA automation and Microsoft Access dashboards, testing fixes and providing feedback to the Data Analyst and Product Engineer.

Air Force Engineer *Mar 2024 – Sep 2025*

- Trained as Flight Line Crew (FLC) and in General Maintenance Tasks (GMT); performed pre- and post-flight checks and supported aircraft maintenance to meet operational readiness requirements

SAP **Singapore**
Machine Learning Engineer Intern *Feb 2021 – Aug 2021*

- Gathered stakeholder requirements, built proof-of-concept solutions, and conducted feature experiments and model evaluations across multiple datasets, presented findings to stakeholders.
- Co-developed an NLP-based autocomplete prototype, structuring the solution into a modular pipeline for data preprocessing, entity extraction, mapping, and search.

Projects

AlphaReGen: Network Security Simulation **GitHub**
Diploma Final Year Project (Grade A) | pfSense, Palo Alto firewall, Wireshark *Oct 2022 – Mar 2023*

- Led the team in developing a simulated attacker–target environment with a traffic generator/responder to simulate network protocols and attack scenarios (e.g. DNS exfiltration, SSH vs. FTP).
- Integrated Wireshark-based traffic analysis and a technical guide to help learners identify attack patterns, distinguish normal from spoofed traffic, and configure and compare pfSense and Palo Alto firewalls.

Leadership & Activities

SUTD Diving Club — Treasurer Mar 2026 – Present
SUTD Innovation Maker Breaker Space - Maker Guide Jan 2026 – Aug 2026
Singapore Polytechnic — Vice-Class President Apr 2021 – Sep 2021

Technical Skills

Programming: Python, JavaScript, VBA, HTML/CSS, SQL (basic)
Libraries & ML: pandas, TensorFlow, Keras, spaCy (NLP)
Development Tools: Microsoft Access, Microsoft Copilot Studio
Security Tools: Wireshark, pfSense, Palo Alto firewalls, Linux (Kali)

Courses & Self-Study

Machine Learning Specialization, DeepLearning.AI, self-paced, in progress